Ben Petrihos
| 703-472-9333 | blpetrihos@wm.edu | U.S. Citizen |

EDUCATION

William & Mary, Williamsburg, VA | Expected Graduation: Spring 2027
Bachelor of Arts in Government, Minor in Data Science | Major GPA: 3.94 | Cumulative GPA: 3.84
VICEROY Scholar, Department of War Cyber-Spectrum Leadership Program | Selected Fall 2026

EXPERIENCE

International Security Research Fellow, Futures Group – Williamsburg, VA | Academic Year 2026–2027
- Recruited by Professor Dennis A.V. Smith as one of four university research fellows; leading a team of two associates conducting a net assessment of U.S. and PRC regional competitions
- Producing international security deliverables for U.S. national security officials, focused on five-to-ten-year strategic scenarios

Research Fellow, Geopolitics of Technology Initiative – Williamsburg, VA | Academic Year 2026–2027
- Leading a team on behalf of the Global Research Institute, for the W&M / Yale policy partnership on the impact and direction of emerging technology on geopolitical competition
- Principal author of a yearlong research effort culminating in publication and presentation at national security conferences

Founding Fellow and White Cell Lead, Statecraft Simulations Group – Williamsburg, VA | 2025 – Present
- Managing the design and operations of wargames focusing on soft power, homeland security, and U.S.–China competition, including AI-modeled economic, national interest, and escalation adjudication
- Collaborated with National Defense University, J7, and J8 to develop tools and analyze simulation findings
- Presented outcomes to policymakers, senior U.S. officials, and private sector leaders
- Selected as a founding fellow under the mentorship of Major General Charlie Lyon, USAF (Ret.)

Data Science Independent Study, William & Mary – Williamsburg, VA | Fall Semester 2026
- Developing deterministic Python agents that aid in adjudication of national security simulations for academic credit, on behalf of the Chair of the Data Science Department, Professor Dan Runfola
- Serving as a research assistant to faculty on Python agent pipelines used in senior seminar adjudication
- Instructing students in how to design and build agents for national security analysis

Research Fellow, Robert M. Gates Initiative in American Statecraft – Williamsburg, VA | 2025– Present
- Developed economic, diplomatic, and information software tools to measure U.S. and partner exposure to Chinese rare earth leverage across extraction, refinement, and manufacturing
- Built an ArcGIS interactive dashboard of 32 critical minerals, including six rare earths, mapping 1,312 supply chain links across 109 countries, with cascading selectors and a 22-country map for national security simulations
- Appointed by Eric Brown, Senior Research Scientist, Global Research Institute

Department of War, Innovation for Defense – Beale AFB, CA | Langley AFB, VA | Fall Semester 2025
- Led a team tasked with developing a base-level counter drone (C-sUAS) effort: conducted 94 stakeholder interviews and site visits at Langley and Beale AFB with Security Forces, Operational Support Squadron, Explosive Ordnance Disposal, and Air Traffic Control
- Root cause analysis resulted in the stand-up of Beale AFB's first C-sUAS Common Operating Picture; identified capability gaps that drove immediate procurement of multiple sensors for layered defenses and laid the groundwork for validating a new drone-defeat framework
- Recruited and mentored by Major General Mark Matthews, USAF (Ret.)
- Identified as team leader by sponsor Captain Quinten Williams (9th OSS Beale AFB)

Research Analyst, American Legislative Exchange Council (ALEC) – Crystal City, VA | 2022

• Conducted data analysis and produced research deliverables for the Economic, Tax, and Public Affairs departments, contributing to multiple ALEC publications

• Represented ALEC at national policy conferences, briefing research to legislators and policy staff, and was credited in published research for analytic contributions

PROJECTS

- **Statecraft Simulations Group: The Petrihos Lever Index (PLI)**
  - Economic statecraft adjudication model for national-security simulations: a tailored AI integrated system whose deterministic engine computes quarterly macroeconomic indicator paths
  - Continuing development in Python of PLI and SDK agent pipelines for White Cell and wargaming use, including support to faculty as a research assistant and instruction of students in agent design for national security objectives
  - Automates economic, national interest, and escalation adjudication for White Cell functionality using JSON Schema worksheets, GitHub Actions, and a third-party cloud server
- **Geopolitics of Technology Initiative: The PRC Economic Vulnerability Index (PRCVI)**
  - Weighted composite indicator of partner-state vulnerability to Chinese economic statecraft, combining chokepoint-adjusted critical-input dependence, concentrated FDI exposure, opaque PRC lending, and services-channel dependence
  - Built a Python pipeline in pandas, NumPy, scikit-learn, and SciPy from AidData TUFF 3.0, World Bank, and UN Comtrade, with a validation harness scored against the Reynolds-Goodman CSIS inventory of PRC economic coercion cases
  - Built an interactive map with hover tooltips and click-to-select inspection, equal-radius markers, and layers for Vulnerability, CCD, HAE, OLE, SCD, plus Reynolds-Goodman coercion attempts
- **Department of War: CTRL, ALT, Defend – Base-Level Counter Drone Readiness**
  - Root cause findings supported stand-up of Beale AFB's first C-sUAS Common Operating Picture and immediate procurement of multiple sensors for layered defenses
  - Designed a joint OSS/SF training framework with pass/fail metrics for detection and response time, and delivered a Mission Model Canvas and deployment roadmap to 9th OSS
- **American Legislative Exchange Council: Unaccountable and Unaffordable**
  - A state pension liability report surveying 290+ state-administered public pension plans across all 50 states (FY 2011–2018)
  - Compiled plan-level assets, liabilities, and funding metrics used to estimate nearly $5 trillion in risk-free unfunded liabilities (~$15,080 per capita) and produced 50-state rankings

LETTERS OF RECOMMENDATION

- Maj Gen Charlie Lyon, USAF (Ret.)
- Maj Gen Mark T. Matthews, USAF (Ret.)
- Capt Quinten E. Williams, USAF
- Skip Estes, Legislative Manager, ALEC Center for Fiscal Reform
- Thomas Savidge, Research Manager, ALEC Center for Fiscal Reform

DEPARTMENT OF THE AIR FORCE
9TH OPERATIONS SUPPORT SQUADRON (ACC)
BEALE AIR FORCE BASE, CALIFORNIA

30 July 2026

MEMORANDUM FOR RECORD

SUBJECT: Letter of Recommendation for Ben Petrihos

1. It is my distinct pleasure that I offer my strongest recommendation for Mr. Ben Petrihos. I had the privilege of serving as his project sponsor for his Hacking 4 Defense team, a program that tasks talented students with solving real-world national security challenges.

2. His team was tasked with identifying how the Air Force could replicate realistic small Unmanned Aircraft System (sUAS) threats to enable effective training and the evaluation of base-wide counter-sUAS (C-sUAS) plans. In tackling this, I watched as Mr. Petrihos led his team through an iterative problem-solving cycle. He conducted dozens of stakeholder interviews, validated operational assumptions and frameworks, and performed a deep root cause analysis of the problem set.

3. Ben's leadership was particularly evident during a site visit to Beale AFB. He immediately distinguished himself by leading discussions with various base agencies and asking tough, insightful questions. He used his ability to map and identify the gaps between strategic guidance, organizational policy, unit resourcing, and front line operations to lead his team in developing a structured training and evaluation program for C-sUAS operations at Beale AFB.

4. His team's root cause analysis directly led to the stand up of Beale AFB's first ever C-sUAS Common Operating Picture, a strategic win for numerous stakeholders that his team worked with. Additionally, the capability gaps his team identified resulted in the immediate procurement of multiple sensors to create layered defenses and laid the groundwork for validating a new drone-defeat framework.

5. Mr. Petrihos is a strong team player and critical thinker who will be an outstanding asset to any organization. He has my highest and unreserved recommendation. Should you have any questions, please contact me at quinten.williams.1@us.af.mil or on my cell at 740-407-7631.

QUINTEN E. WILLIAMS, Capt, USAF
Flight Commander

William & Mary
Arts & Sciences

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Aug 3, 2026

Letter of Recommendation for Benjamin "Ben" Petrihos
To Whom It May Concern

After 34 years on active duty and serving with Raytheon as VP for US Air Force programs, I was given the opportunity to teach the "Innovation for Defense (I4D)" course at the College of William & Mary. Ben is one of the 130+ students I've now had the pleasure to guide since 2021. Sponsored by the US Department of Defense's Defense Innovation Unit (DIU), active-duty military and Intelligence Community sponsors submit "Challenges" -- essentially gnarly problems -- that our student teams seek to solve using an entrepreneurial approach in delivering a working Minimum Viable Product by semester's end. Fifty years of experience across air, space, and cyber missions -- in uniform, in industry, and now in the classroom -- have given me a reliable eye for talent. Ben has it.

Challenged by Airmen at Beale AFB in California, Ben's team tackled the growing threat to base-level readiness posed by small, unmanned aircraft. Ben conducted nearly 100 interviews with Security Forces, Explosive Ordnance Disposal, Air Traffic Control, and Operational Support personnel at Langley and Beale Air Force Bases. That deep dive exposed integration gaps in base defense between USAF base security forces, who traditionally held the mission, and members of the operational support squadron newly charged with this mission area. With a clear-eyed understanding of what was wrong, Ben and his team turned what Airmen told him into a joint training framework with pass/fail metrics a squadron could put to use.

What I value most in Ben is the speed at which his ideas become working tools. As a founding fellow in the Statecraft Simulations Group, part of former Secretary of Defense Robert Gates' Initiative for American Statecraft, he built the Petrihos Lever Index, an AI-integrated model that automates economic and escalation-risk adjudication for wargame White Cells. When existing gaming platforms proved inadequate, he wrote his own: a live exercise environment with role-based play, synchronized session control, automated adjudication, and full operator traceability, replacing the improvised chat channels such exercises too often run on. He then added a Data Science minor and arranged an independent study to keep developing the platform. Ben accomplishes in days what takes others weeks.

I have spent much of my career developing officers and engineers for the joint fight in air and space, land and sea. Those who excel have common traits: curiosity, creativity, and tenaciousness -- they care about what they do. But most importantly, they are leaders who can unify many in common purpose.

Of those 130+ students I've had the honor to lead and mentor, Ben is in the top five. He has my strongest endorsement and a clean slate in front of him waiting to be filled with our country's greatest challenges.

Mark T. Matthews
Major General, USAF (retired)
Adjunct Lecturer

WILLIAM & MARY

Dear Sir or Madam,

I strongly endorse Ben Petrihos. After 32 years on active duty, including service as Chief of the Joint Staff's Studies, Analysis, and Gaming Division (J-8) and as Director of Operations at Air Combat Command, I now teach wargaming at William & Mary and lead the Statecraft Simulations Group.

I met Ben when he enrolled in my "Wargaming: From Theory to Practice" course. He was the top student performer in that course, distinguished by thoughtfulness, rigor, and analytic skill. The curriculum included wargaming theory and applications from the Naval War College, Air War College, and National Defense University. Peter Perla's writings were a core part of our design and analysis work.

Ben has visited the Joint Forces Staff College and the Joint Staff/J-7 wargaming branch several times, where he learned matrix-style and seminar-style wargaming. He served as one of two design leads for an internally developed matrix game that examined United States-Russia strategic competition as it unfolds in the Arctic region.

Ben impressed me with his maturity, depth of analysis, and intellectual curiosity. He sought additional game design readings and game reports to deepen his knowledge, always with an eye on analysis. I hand-selected Ben as an initial cadre fellow of our Statecraft Simulations Group, part of former Secretary of Defense Robert Gates' Initiative for American Statecraft. The group advocates for the use and expansion of "soft power" through research and seminar-style gaming. Its aim is to explore the nation's most pressing national security challenges and to educate policymakers and the public on the rising risks associated with the People's Republic of China.

Ben now serves as White Cell Lead and economics subject matter expert. He designed the Petrihos Lever Index, a manually coded, AI-integrated model whose deterministic engine computes macroeconomic indicator paths. The White Cell uses it to adjudicate economic effects, national interest, and escalation through a recorded, repeatable method grounded in peer-reviewed sources. That model is now part of how we run our simulations and will remain in use long after he leaves. He has briefed policymakers, administration officials, and private sector leaders. He has also been awarded a Data Science independent study to keep developing these tools for my senior seminar.

Ben has my strongest recommendation. Select him, and you will be glad to have him on your team.

Sincerely,



CHARLES W. LYON
Major General, U.S. Air Force (retired)
Adjunct Lecturer, Government Department

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